

Self-taught AI builder — learning from papers and courses since early 2026, I designed, built and evaluated **11 projects in three months**, spanning training-method research, multi-device AI systems, applied decision systems and shipped products.

SELECTED PROJECTS

Zero Gradient — 4.16B-param LM trained with no backpropagation RESEARCH
Trained a content-routed MoE LM with autograd globally disabled on one T4; a 5-phase study + controlled experiment proved the capability ceiling is architectural, not algorithmic. Paper draft + arXiv package.
PyTorch · MoE · layer-local learning · WikiText-103

Gemma4all — cross-device AI task system SYSTEMS
Schema-first (42 JSON schemas), event-sourced, local-first system; fail-closed human-in-the-loop approvals and a privacy-based routing engine across iOS/Android/macOS; on-device inference via LiteRT-LM on Metal GPU.
TypeScript · Python · Swift · Kotlin · event sourcing

TriageGeist — trustworthy ED-triage decision stack APPLIED ML
Found 99.7% label leakage and refused to exploit it; built a text-blind GBDT (QWK 0.93) + isotonic calibration + conformal uncertainty + bootstrap fairness audit + ops console. 12/12 tests.
LightGBM · conformal prediction · calibration · fairness

NeuroGolf (ARC-AGI) — minimal ONNX circuit compiler RESEARCH
Reframed a "neural golf" contest as program synthesis; reverse-engineered the scorer into a cost model and compiled true task rules into overfitting-proof integer-weight circuits validated on 3000+ samples.
ONNX · program synthesis · scorer reverse-engineering

Nemotron Post-training — post-training pipeline, 30B MoE reasoning model RESEARCH
Reproducible rule-teacher → CoT-SFT pipeline (EM 0.75; 30B MoE in 4-bit on one GPU, upstream cache bug patched, reward shaping); root-caused the training failure across six postmortems and designed the recovery route.
Transformers · PEFT/LoRA · 4-bit · GRPO reward shaping

AI Showrunner — one sentence → subtitled vertical short film PRODUCT
Multimodal agent pipeline (script → storyboard → video generation) with a vision-model critic loop driving targeted retries and cross-shot character consistency; six end-to-end films produced.
FastAPI · Qwen-VL critic · I2V/T2V · ffmpeg

Also built: Offline French Companion (in-browser LLM tutor — WebLLM + Whisper + OCR, fully offline) · Orbit Wars (multi-agent evaluation methodology) · Maze Crawler (strategy agent + CI) · VoiceInput (pure-Swift macOS speech input) · Atelier Post (shipped art-to-Instagram tool, web/PWA/iOS).

EXPERIENCE

BNP Paribas · Global Corporate Banking
Paris · Dec 2025 — Present
Cross-border transactional & financing solutions for multinational clients; built internal RAG knowledge bases (Open WebUI / Dify) for client-response efficiency.

HSBC · Global Markets, Liquidity
Paris · Jun — Nov 2024
Optimized short-term EMEA portfolios under LCR/NSFR; built VBA/Excel pricing & rate-tracking tools; contributed to the ALCM committee.

TikTok · Growth Strategy
Beijing · Feb — May 2023
10+ strategic projects (+23% SaaS/AI market share); SQL/Excel analytics for a 20+ team; +13% user conversion in one month.

Uber China · DiDi · Product & User Ops
Beijing · Aug 2022 — Jan 2023
3 business-operations projects; Python/SQL/VBA analysis and predictive models for data-driven user strategies.

EDUCATION

ESCP Business School 2023 — 2026
MSc in Management, Finance · Paris & London.

Beijing Intl Studies Univ. 2019 — 2023
BA Language & Culture (French, Japanese); minor New Media (CUC); self-taught Psychology (Yale/Coursera).

SKILLS

Programming: Python · TypeScript/Node · Swift · Kotlin · JS · SQL · VBA

AI / ML: PyTorch · LoRA/PEFT · MoE · calibration · conformal prediction · LiteRT-LM · ONNX · WebLLM · Whisper · LLM/agent orchestration · schema-first & event-sourced systems

Languages: Chinese (native) · English · French (fluent) · Japanese (working)

Strengths: high agency, self-taught zero-to-one builder — curious, reflective and resilient, with strong insight and a painter's eye, at home across cultures.